Ripple Tank

Exam code: 0625 & 0972

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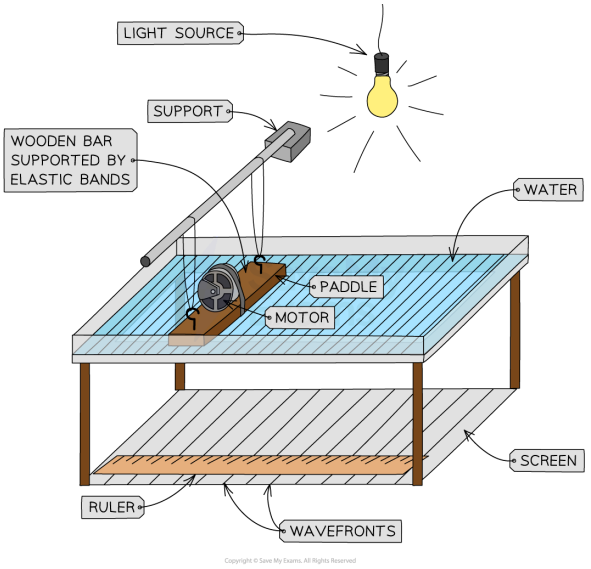
Investigating waves with a ripple tank

- Ripple tanks are commonly used in experiments to demonstrate the following properties of water waves:
  - **Reflection** at a plane surface
  - **Refraction** due to a change in speed caused by a change in depth
  - **Diffraction** due to a **gap**
  - **Diffraction** due to an **edge**

Ripple tank experimental set-up



Reflection, refraction and diffraction can be demonstrated using a ripple tank

- **Wavefronts** from the transverse water surface waves can be viewed and analysed on the screen illuminated to show below the tank

Investigating Reflection

- **Wavefronts** are reflected off a metal bar (plane surface) placed in the water of the ripple tank
- When the bar is placed at an angle to the wavefronts of the waves generated by the paddle reflect according to

Help me understand

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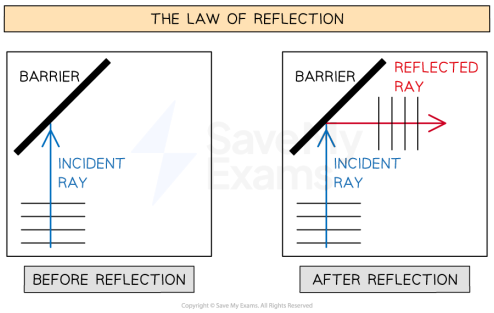
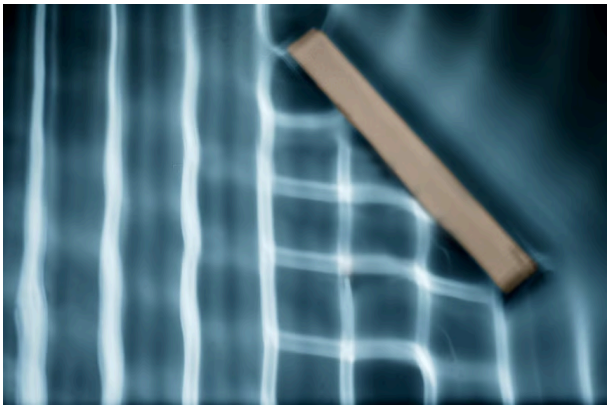
Ask questions and get help with exactly what you're revising.
Got it

ripple tank



Incident wavefronts are reflected at 90 degrees against a barrier

Reflected wavefronts in a ripple tank

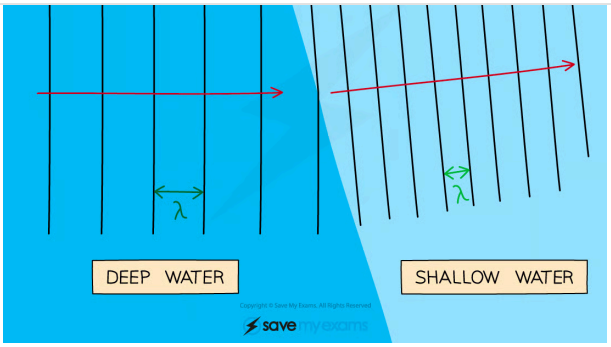


Wavefronts of incident and reflected waves form right angles to each other

Investigating refraction

- Refraction can be shown by placing a **glass block** in the tank
  - The glass block should sit below the surface of the water and cover only some of the tank floor
- The depth of water becomes **shallower** where the glass block is placed
  - Since speed depends on depth, the ripples slow down when travelling over the block
- The water surface waves **slow down** when passing from **deep to shallow** water in the ripple tank

Refracted wavefronts in a ripple tank

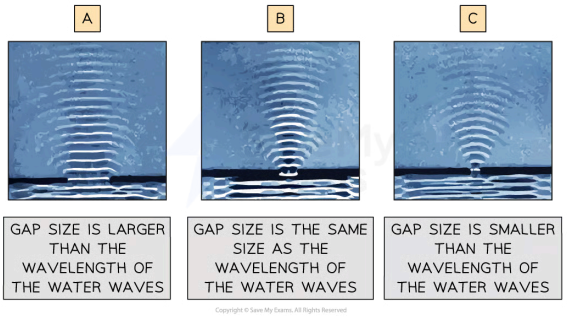


When water waves travel from deep areas to shallow areas they slow down

Investigating diffraction

- **Diffraction** can be shown in a ripple tank by placing small barriers with a **gap** or an **edge** in the tank
- The amount of Diffraction that occurs can be changed by changing the wavelength of the waves compared to the gap size

Changing the gap size for diffraction in a ripple tank

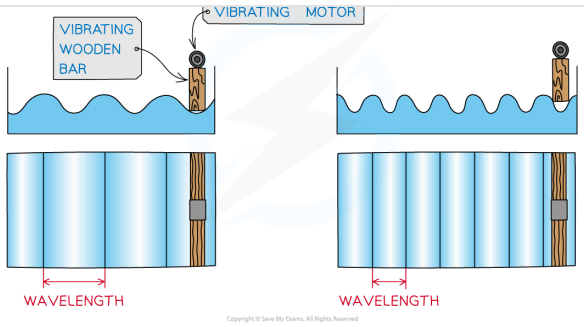


When the gap size is bigger than the wavelength less diffraction occurs and the waves spread out less after passing through

Changing the wavelength of waves in the ripple tank

- The **motor** creates the up-and-down movement of the **paddle**
- The frequency of the motor affects the wavelength of the waves generated by the paddle
- The diagram below shows how the wavelengths differ with frequency in a ripple tank
  - The **higher** the frequency of the motor, the **shorter** the wavelength
  - The **lower** the frequency of the motor, the **longer** the wavelength

Wavelength and frequency of waves in a ripple tank



Ripple tank patterns for low and high-frequency vibration

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